

PROJECT REPORT

DESIGN AND FABRICATION OF AN AUTONOMOUS AI-DRIVEN CATAMARAN FOR REAL-TIME SURFACE DEBRIS COLLECTION AND AQUATIC MONITORING

Project Platform: ECO-FLOAT (Version 2.0)

CHAPTER 1: INTRODUCTION & PROBLEM STATEMENT

1.1 Introduction

The preservation of urban aquatic environments—such as lakes, public ponds, and reservoirs—is increasingly threatened by the unchecked accumulation of anthropogenic solid waste. Macroplastics, polystyrene containers, and synthetic packaging form dense floating layers on stagnant or slow-moving water surfaces. Unlike open marine systems, urban inland water bodies lack strong, continuous currents to disperse these pollutants, leading to immediate localized ecological degradation.

1.2 Problem Statement

Traditional approaches to surface water remediation suffer from significant operational bottlenecks:

- **Manual Labor Inefficiencies:** Relying on human operators deploying manual skimming nets from shorelines or manned boats is slow, physically taxing, and costly over long periods.
- **High Capital/Power Costs:** Advanced remote sensing platforms often utilize heavy, power-hungry active conveyor systems or complex hyperspectral radiometer arrays. These systems require highly specialized parts, drain batteries rapidly, and are cost-prohibitive for widespread local deployment.
- **Blind Pathing:** Basic automated skimmers typically traverse pre-programmed paths blindly, expending valuable battery energy clearing clear water rather than targeting dense waste zones.

1.3 Project Objective

The **ECO-FLOAT** project introduces a lightweight, low-cost Unmanned Surface Vessel (USV) that re-engineers autonomous water cleaning. By merging a highly stable passive catamaran hull with edge-computed computer vision, the platform actively searches for, identifies, and targets floating debris. This eliminates the need for heavy active collection machinery or complex external sensor arrays, providing an accessible, continuous, and highly targeted remediation solution.

CHAPTER 2: MECHANICAL DESIGN & HYDRODYNAMICS

The mechanical framework of ECO-FLOAT is engineered to maximize passive physical sorting while minimizing active component failures.

2.1 Catamaran Hull Architecture

The vessel utilizes a twin-hull (catamaran) configuration. This choice is driven by the necessity for high stability on unpredictable, shifting fluid surfaces.

- **Metacentric Stability:** Spacing two separate asymmetric hulls across a wider beam width dramatically increases the vessel's metacentric height (GM). This ensures that as the rear mesh net fills with heavy, water-logged debris, the vessel resists pitching forward or rolling along its longitudinal axis.
- **Asymmetric Entry Profile:** The inner walls of the dual hulls are relatively flat, while the outer walls are hydrodynamically curved. This asymmetry prevents water from packing tightly at the bow, encouraging fluid to slice cleanly around and through the vehicle.

2.2 Passive Induction Channel & Funneling

Instead of using motorized conveyor belts—which are prone to jamming when water weeds, aquatic moss, or thin plastic bags wrap around rotating gears—ECO-FLOAT uses fluid dynamics to trap waste.

- **Converging Nozzle Design:** The forward interior sections of the twin hulls form a wide entry mouth that narrows toward the midsection.
- **Velocity Differential:** As the vessel moves forward at a steady operational velocity (v), water entering the wide bow channel is squeezed into a narrower cross-sectional area. According to the principle of continuity, this reduction in area creates a localized increase in fluid velocity relative to the hull, sweeping floating plastics cleanly backward into the containment zone without auxiliary power.

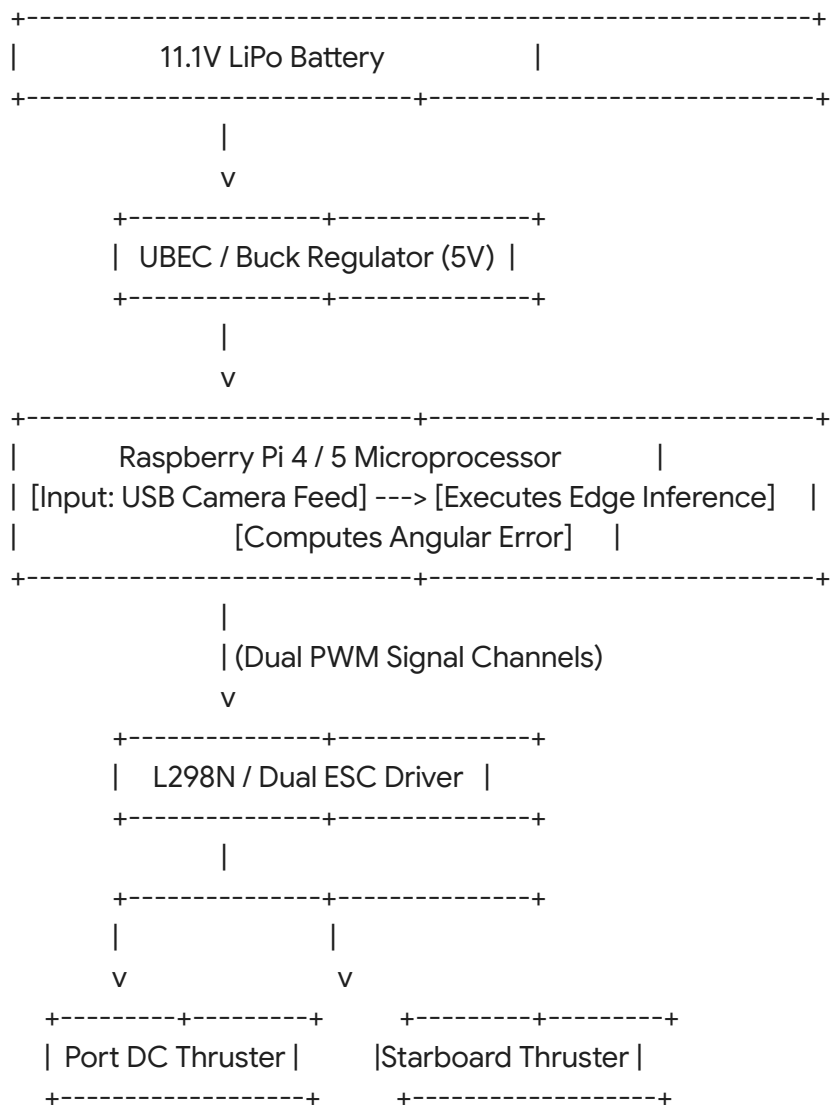
2.3 Trailing Mesh Containment

At the rear termination point of the central induction channel sits a rigid, lightweight frame holding a collection net.

- **Material Selection:** Built from High-Density Polyethylene (HDPE) mesh, which is naturally hydrophobic. This prevents the net material itself from absorbing water and bloating vessel weight.
- **Pore Optimization:** A mesh pore size of approximately 5 mm balances two conflicting engineering constraints: it provides ample surface area for water to vent freely out the back (minimizing trailing hydrodynamic drag), while securely trapping microplastics, cups, snack wrappers, and bottles.

CHAPTER 3: ELECTRONICS & EMBEDDED SYSTEM ARCHITECTURE

The electronics bay is sealed within a waterproof center-deck compartment, optimizing power routing and minimizing signal-to-noise ratios across the edge compute stack.



3.1 Hardware Configuration Details

- **Primary Processor:** Raspberry Pi single-board computer. This serves as the centralized edge-computing node, handling concurrent tasks including video stream decoding, tensor manipulation, and motor output generation.
- **Optical Transducer:** A fixed-focus USB camera module equipped with a 120° wide-angle lens. It is hard-mounted to an elevated forward mast, angled downward at roughly 15° to monitor both long-range targets on the horizon and imminent obstructions near the bow.
- **Actuation & Propulsion:** Dual 12V high-torque underwater DC marine thrusters mounted to the rear transom of each hull. Steering is completely differential; the platform does not use a mechanical rudder, eliminating a major physical point of failure.
- **Power Subsystem:** A high-capacity 11.1V Lithium-Polymer (LiPo) battery pack channels raw current directly to the motor drivers. A step-down Buck Regulator (UBEC) stabilizes a clean $5\text{V}/3\text{A}$ rail to power the Raspberry Pi without risk of voltage sagging during high thruster acceleration.

CHAPTER 4: ARTIFICIAL INTELLIGENCE & SOFTWARE LOGIC

ECO-FLOAT completely avoids the need for heavy cloud computing or cellular data links by confining its computer vision pipeline to localized hardware.

4.1 Edge Vision Pipeline

The software subsystem continuously processes video frames through a highly optimized, lightweight neural network architecture (e.g., a localized custom object detection model like YOLOv8-nano or MobileNet-SSD).

1. **Frame Acquisition:** The camera inputs raw frames at a resolution of 640×480 pixels to preserve processing speed.
2. **Tensor Inference:** The onboard model screens the imagery, actively filtering out environmental noise unique to water surfaces, such as shimmering solar glare, moving ripples, and natural organic matter (duckweed, fallen leaves).
3. **Target Bounding:** When an artificial plastic object crosses a confidence threshold ($\geq 75\%$), the system generates a dynamic bounding box and extracts its horizontal center coordinate (X_c).

4.2 Proportional Control Steering Logic

To transform visual detections into kinetic movement, the system calculates the pixel distance between the object's center (X_c) and the static center-line of the camera's frame ($X_{center} = 320$).

The steering error is defined as:

$$\Delta e = X_c - X_{center}$$

This pixel error scales motor power through a proportional control feedback loop:

- **Case 1: Direct Intercept ($\Delta e \approx 0$)**
The plastic target is dead center. The Raspberry Pi sends equal PWM duty cycles to both the port and starboard motors, accelerating forward smoothly.
- **Case 2: Starboard Deviation ($\Delta e > 0$)**
The plastic object is floating off to the right side of the vessel. The system scales up the PWM duty cycle to the **port motor** while decreasing power to the **starboard motor**. The resulting thrust imbalance swings the bow smoothly to the right, re-centering the target object in the lens frame.
- **Case 3: Port Deviation ($\Delta e < 0$)**
The object is to the left. The **starboard motor** increases output while the **port motor** scales back, correcting the vessel's trajectory to the left.

CHAPTER 5: COMPLETE OPERATIONAL WORKFLOW

The entire mission life cycle of the ECO-FLOAT platform operates within a closed, autonomous logical loop, categorized from launch to safe recovery.

1. System Initialization & Built-In Test:Phase 1.

The platform is placed in the water. Flipping the primary breaker boots the Linux core on the Raspberry Pi. A diagnostic script verifies video input streams, checks battery voltage rails, and cycles both thrusters at low RPM to ensure electrical continuity.

2. Exploratory Search Protocol:Phase 2.

With no initial debris targets detected, the bot executes a pre-mapped exploratory lawnmower or zig-zag path across the water body. It monitors its heading via an onboard digital compass to guarantee full surface area coverage.

3. Visual Target Lock:Phase 3.

The vision model identifies a floating plastic bottle with high confidence. The exploratory grid is immediately suspended; the system initializes its tracking thread and locks onto the bounding box coordinates.

4. Closed-Loop Interception Navigation:Phase 4.

The proportional steering logic continuously alters individual motor speeds based on tracking error. The catamaran changes course dynamically, aligning its central funnel channel directly with the drifting debris.

5. Passive Capture & Recovery Enclosure:Phase 5.

As the bot makes contact, the forward momentum pushes the bottle through the converging catamaran hulls. Fluid pressure slides it into the trailing HDPE mesh net where it remains trapped. The system clears the target lock and returns to Phase 2 to hunt for more waste.

CHAPTER 6: ADVANTAGES & SYSTEM COMPARISONS

Evaluating the operational parameters of this camera-based configuration against older or more complex alternative methodologies highlights several clear design advantages:

Engineering Parameter	Traditional Active Systems	Hyperspectral / Sensor Bots	ECO-FLOAT (Camera-AI Model)
Mechanical Complexity	High (Motors, active gears, conveyors)	Low (Static hull structures)	Low (Zero moving collection parts)
Power Requirements	Very High (\$200\text{W}+\$ for conveyor lift)	High (\$50\text{W}+\$ for radiometer arrays)	Very Low (\$15\text{W}\$-\$25\text{W}\$ total consumption)
Vulnerability to Weeds	Severe (Weeds instantly jam rotating gears)	None (Purely observational)	None (Passive channel avoids entanglement)
Path Planning Style	Blind / Non-targeted sweeping paths	Highly complex data analysis mapping	Reactive / Direct optical interception
Manufacturing Cost	High (Heavy mechanical assemblies)	Extremely High (Specialized optical sensors)	Very Low (Uses accessible components)

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The updated **ECO-FLOAT** platform successfully transitions surface water remediation away from expensive, fragile sensor arrays and power-heavy active lifting gear. By utilizing a highly stable catamaran hull coupled with edge-computed computer vision, the platform delivers a reliable, targeted, and highly energy-efficient solution. Running the AI model entirely on localized hardware ensures the bot functions perfectly in remote or network-isolated urban reservoirs, offering a highly scalable blueprint for automated environmental cleanup.

7.2 Future Scope

- **Solar Array Integration:** Mounting semi-flexible photovoltaic panels across the top deck of both catamaran hulls would allow for continuous trickle-charging, significantly extending deployment windows.
- **Swarm Coordination:** Configuring multiple ECO-FLOAT vessels to communicate via low-power mesh networking (such as LoRa modules) would enable them to divide large lakes into collaborative cleaning sectors, drastically increasing total daily waste collection volume.